

```

arr = [1,2,3,4,5] #1d
print(type(arr))

arr1 = [[1,2],[3,4],[5,6]]
print(type(arr1))

print(arr + arr1)

arr[1] = 10
arr

arr[3:] = [40,50]
arr + arr1

arr1[0][1] = 20
arr1[1][1] = 30
arr1

```

```

<class 'list'>
<class 'list'>
[1, 2, 3, 4, 5, [1, 2], [3, 4], [5, 6]]
[[1, 20], [3, 30], [5, 6]]

```

```

a = list(map(int,input().split()))
print(a)
print(type(a))

```

```

1 2 3
[1, 2, 3]
<class 'list'>

```

```

import array as arr
a = arr.array('i',[1,2,3,4,5])
print(a)
print(len(a))

```

```

#"i" Integer
#"f" float
#"d" double
#"u" unicode characer

```

```

array('i', [1, 2, 3, 4, 5])
5

```

```

#input from user, arr input of numbers
#print sum of numbers in that array

#with using numpy
import numpy as np
a = np.array(list(map(int,input().split())))
print(np.sum(a))

#without using numpy
b = list(map(int, input().split()))
total_sum = 0
for num in b:
    total_sum += num
print(total_sum)

```

```

2 31 2
35
3 4 5
12

```

```

#min and max of an array

# numpy
import numpy as np
a = np.array(list(map(int,input().split())))
print(np.min(a))
print(np.max(a))

```

```
1 2 3
1
3
```

```
b = list(map(int, input().split()))
print(max(b))
print(min(b))
```

```
1 2 3
3
1
```

```
#without inbuilt
c = list(map(int, input().split()))
mini = maxi = c[0]
for num in c:
    if num > maxi:
        maxi = num
    if num < mini:
        mini = num
print(mini)
print(maxi)
```

```
1 2 3
1
3
```

```
a = [3,2,4,2,5,6]
print(sorted(a))
print([a[-2]])
```

```
[2, 2, 3, 4, 5, 6]
[5]
```

```
#2nd largest
a = list(map(int, input().split()))
for i in range(0, len(a)):
    for j in range(i+1, len(a)):
        if a[i] < a[j]:
            a[i], a[j] = a[j], a[i]
print(a[1])
```

```
12 32 11
12
```

```
a = list(map(int, input().split()))

largest = float('-inf')
second = float('-inf')

for num in a:
    if num > largest:
        second = largest
        largest = num
    elif num > second and num != largest:
        second = num

print(second)
```

```
33 2 1 34
33
```

```
nums = [2, 2, 1, 4, 5, 1, 6, 6, 6, 4]
seen = {}
for num in nums:
    if num not in seen:
        seen[num] = True
print(sorted(seen))
```

```
[1, 2, 4, 5, 6]
```

```
n = int(input())
result = 1
for i in range(n, 1, -1):
```

```
result = result * i
```

```
5
120
```

```
# count each number
nums = [2, 2, 1, 4, 5, 1, 6, 6, 6, 4]
snums = sorted(nums)
count = {}
for num in snums:
    if num not in count:
        count[num] = 1
    else:
        count[num] += 1
print(count)
```

```
{1: 2, 2: 2, 4: 2, 5: 1, 6: 3}
```

```
import numpy as np
list1 = np.array([1, 2, 4])
list2 = np.array([1, 3, 4])
list3 = np.concatenate((list1, list2))
list3 = np.sort(list3)
list3 = list3.tolist()
print(list3)
```

```
[1, 1, 2, 3, 4, 4]
```

```
class Solution:
    def twoSum(self, nums, target):
        for i in range(len(nums)):
            for j in range(i + 1, len(nums)):
                for k in range(j + 1, len(nums)):
                    if nums[i] + nums[j] + nums[k] == target:
                        return [i, j, k]
```

Enumerate() is a python function that gives you both the index and the value while looping through a list

```
# 1. Define your input data first
nums = [2, 7, 11, 15]
target = 9
```

```
# 2. Run your Two Sum logic
seen = {}
for i, num in enumerate(nums):
    complement = target - num
    if complement in seen:
        print(seen[complement], i)
        break
    seen[num] = i
```

```
0 1
```

```
#12345
#45123

a = [1,2,3,4,5]
b = a + a

print(b[3:8])
```

```
[4, 5, 1, 2, 3]
```

```
arr = [1,2,3,4,5]
k = int(input("k: "))
n = len(arr)
arr1 = [0] * n
for i in range(n):
    arr1[(i+k)%n] = arr[i]
print(arr1)
```

```
k: 2
[4, 5, 1, 2, 3]
```

```
a = [2, 5, 8, 9]
b = [3, 4, 7, 8]

c = [0] * (len(a) + len(b))

i = j = k = 0

while i < len(a) and j < len(b):
    if a[i] <= b[j]:
        c[k] = a[i]
        i += 1
    else:
        c[k] = b[j]
        j += 1
    k += 1

while i < len(a):
    c[k] = a[i]
    i += 1
    k += 1

while j < len(b):
    c[k] = b[j]
    j += 1
    k += 1

print(c)
```

```
[2, 3, 4, 5, 7, 8, 8, 9]
```

```
a = [2, 5, 8, 9]
b = [3, 4, 7, 8]

c = [0] * (len(a) + len(b))
i = j = k = 0

while i < len(a) and j < len(b):
    if a[i] < b[j]:
        c[k] = a[i]
        i += 1
    elif a[i] > b[j]:
        c[k] = b[j]
        j += 1
    else:
        c[k] = a[i]
        i += 1
        j += 1
    k += 1

while i < len(a):
    c[k] = a[i]
    i += 1
    k += 1

while j < len(b):
    c[k] = b[j]
    j += 1
    k += 1

print(c[:k])
```

```
[2, 3, 4, 5, 7, 8, 9]
```

```
a = [2, 5, 8, 9]
b = [3, 4, 7, 8]

visited = [0] * len(b)
c = [0] * min(len(a), len(b))
k = 0

for i in range(len(a)):
    for j in range(len(b)):
        if a[i] == b[j] and visited[j] == 0:
```

```
        c[k] = a[i]
        k += 1
        visited[j] = 1
        break

# trim output
result = c[:k]
print(result)  # [8]
```

```
[8]
```